

UNIVERSITÀ DEGLI STUDI DI MILANO-BICOCCA

DATA SCIENCE LAB FOR SMART CITIES

FINAL ESSAY

Wildfire Risk and Fuel Management in Portugal: Spatial Coverage Gaps, Economic Vulnerability, and Policy Interventions

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Abstract

Wildfires represent a major environmental and societal challenge in Portugal, with climate change increasing the frequency and severity of fire-prone conditions. Effective prevention requires not only the identification of high-risk areas but also the equitable allocation of protective infrastructure across territories with different economic capacities. This study evaluates how effectively Portugal's Primary Fuel Management Network (RPFGC) covers high wildfire risk zones and investigates whether municipalities with lower purchasing power face disproportionate prevention coverage gaps.

Using a Data Science and Smart Cities approach, the analysis integrates wildfire risk data, fuel management infrastructure, municipal boundaries, and purchasing power indicators through Python-based geospatial and statistical methods. A composite wildfire risk model is constructed from five spatial layers and classified into five risk levels, identifying a total high-risk area of approximately 4,327 km² across mainland Portugal.

Results show that only 38.6% of high-risk territory falls within the 1 km zone of influence of the RPFGC, leaving approximately 2,656 km² unprotected. Two complementary analyses were conducted to assess whether prevention gaps are concentrated in economically vulnerable municipalities. The first examines the *relative* Gap Ratio: the percentage of each municipality's high-risk area that is unprotected, and finds no evidence of disproportionate concentration in vulnerable municipalities ($r = -0.37$, $p < 0.001$). The second examines the *absolute* unprotected area: the total km² of high-risk land without network coverage, and finds a significant positive relationship with vulnerability ($r = 0.30$, $p < 0.001$), meaning that more economically vulnerable municipalities have larger absolute unprotected areas. Together, these results reveal a territorial inequality that is absolute rather than relative in nature: while the network covers a comparable share of high-risk territory across municipalities, the poorest interior municipalities face a far greater absolute scale of unprotected land, and lack the economic capacity to compensate for it.

This research contributes to the Smart Cities framework by demonstrating how geospatial intelligence and data-driven analysis can support the identification of priority municipalities where wildfire risk, prevention gaps, and economic vulnerability intersect, enabling more targeted and evidence-based policy interventions to improve territorial resilience.

AI Declaration

This report was developed with the assistance of generative Artificial Intelligence tools, including ChatGPT (OpenAI) and Claude (Anthropic), to support brainstorming, report structuring, methodological planning, literature exploration, writing refinement, and coding guidance.

The authors were responsible for defining the research objectives, selecting and processing the datasets, designing and implementing the analytical methodology, interpreting the results, and formulating the conclusions and policy recommendations. Outputs generated by AI tools were critically evaluated, adapted where necessary, and used solely as supporting resources throughout the development of the project. The authors assume full responsibility for the accuracy, originality, and integrity of the work presented in this report.

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1 Problem Description and Indicators

1.1 Problem Description

Wildfires constitute one of the most significant environmental and territorial challenges affecting Portugal. The country is particularly vulnerable to wildfire events due to a combination of climatic, environmental, and socio-economic factors, including its Mediterranean climate, recurrent drought conditions, extensive forested areas, land abandonment processes, and vegetation accumulation. Climate change has increased wildfire risk across Europe, particularly in Mediterranean regions where prolonged droughts, heatwaves, and extreme fire weather conditions have become more frequent [1]. According to the IPCC Sixth Assessment Report, Southern Europe is expected to experience increasing temperatures, more frequent and intense heatwaves, and more severe drought conditions, all of which contribute to elevated wildfire risk and longer fire seasons [2]. The structural roots of this vulnerability are also socio-economic in nature: since the 1960s, rural exodus and rapid industrialisation left vast inland areas of northern and central Portugal depopulated and effectively unmanaged, driving the expansion of fire-prone shrublands and eucalyptus and pine plantations [3]. Between 1980 and 2023, wildfires in mainland Portugal burned a cumulative total of approximately 4.99 million hectares: an area equivalent to roughly 54% of the national territory [4]. The 2017 fire season alone set an all-time record with approximately 540,000 hectares burned, 114 fatalities, over 1,100 injuries, and economic losses exceeding USD 1.2 billion, making it the costliest natural disaster in the country's insurance history [5]. More recently, the 2024 fire season burned over 135,000 hectares across central and northern Portugal in a single week [6]. The OECD has estimated that wildfires now affect hundreds of thousands of people annually in Portugal, generate widespread respiratory health impacts, and represent a structural threat to forest ecosystems, soil integrity, and long-term territorial resilience [7].

Prevention, rather than suppression alone, has increasingly been recognised as the most effective strategy for reducing the frequency and severity of wildfires. In this context, Portugal's Primary Fuel Management Network (*Rede Primária de Faixas de Gestão de Combustível*, RPFGC) constitutes the central instrument of national-scale wildfire prevention. Established under Decree-Law No. 82/2021 and the Integrated Rural Fire Management System (SGIFR), the RPFGC is designed to create large-scale horizontal discontinuities across the landscape through the systematic removal or reduction of vegetation along strategic corridors [8]. Each corridor consists of a central forest road of approximately 6 metres, flanked by 10-metre strips cleared of arboreal and shrub cover, and outer 20-metre strips where canopy spacing is maintained above 4 metres [9]. The network is intended to isolate fire outbreaks, protect roads, critical infrastructure and social facilities, reduce average burned area by compartmentalising the landscape, and improve firefighter safety [10]. Its national implementation is funded with 120 million Euros through the Portuguese Recovery and Resilience Plan (PRR, Project C08-i03), targeting the establishment of 21,000 hectares of fuel management strips across continental Portugal [11]. Despite its strategic importance, the implementation of fuel treatments in Portugal has historically suffered from low execution rates, challenges related to fragmented private land ownership, and insufficient targeting of the territories at highest structural risk [12]. These shortcomings highlight the critical need to evaluate whether the infrastructure being deployed is spatially aligned with zones of greatest wildfire risk, and whether its coverage is equitable across municipalities with differing socio-economic capacities.

This study addresses these challenges by adopting a Smart Cities and Data Science approach to evaluate the spatial effectiveness and equity of Portugal's primary prevention infrastructure. Smart city frameworks have increasingly been applied to disaster risk management, leveraging geospatial intelligence, remote sensing, and data-driven methods to support more transparent, evidence-based, and equitable public decision-making [13]. In this context, the integration of wildfire risk modelling with infrastructure coverage analysis and socio-economic indicators represents a meaningful contribution to territorial resilience. The central research question guiding this study is: *How effectively does Portugal's primary fuel management infrastructure cover high wildfire risk zones, are the gaps in prevention coverage disproportionately concentrated in the most economically vulnerable municipalities, and what targeted policy interventions could address these inequalities?* To answer this question, the study pursues three objectives: (1) to assess the spatial alignment between high wildfire risk areas and the coverage of the Primary Fuel Management Network, at the national scale; (2) to analyse coverage gaps at the municipal level and identify the municipalities with the largest proportions of unprotected high-risk territory; and (3) to examine whether municipalities with lower purchasing power exhibit systematically higher prevention coverage gaps, thereby providing evidence of territorial inequalities in wildfire resilience.

1.2 Indicators

To operationalise the research question, four indicators are defined and applied throughout the analysis. Together, they enable a quantitative assessment of wildfire risk exposure, prevention infrastructure coverage, and the socio-economic conditions of municipalities.

Wildfire Risk Index. A composite spatial indicator that quantifies wildfire risk across mainland Portugal on a scale from 1 (very low) to 5 (very high). It is constructed by combining five input layers: conjunctural hazard, historical burned areas, critical fire locations, vegetation index, and aridity index, through a weighted overlay, where each layer is normalised to a common scale prior to combination. The primary fuel management network layer is also sourced from the same institution. For the coverage analysis, only areas classified as risk class 4 (high) or class 5 (very high) are considered.

Coverage Ratio. The proportion of high-risk territory (classes 4–5) that falls within the zone of influence of the Primary Fuel Management Network (RPFGC), defined as a 1 km buffer around the network. A higher value indicates better alignment between prevention infrastructure and high-risk zones.

Gap Ratio. The complement of the Coverage Ratio, quantifying the proportion of high-risk territory that remains outside the buffer and is therefore considered unprotected. It is the primary indicator used in the municipal-level equity analysis to identify territories where prevention coverage is insufficient relative to wildfire exposure.

Purchasing Power. The indicator of economic vulnerability at the municipal level, specifically the per capita income index (IpC), measured relative to the national average (base = 100). The index is inverted and normalised to a 0–1 scale, where 1 represents the most economically vulnerable municipality, and used to examine whether prevention coverage gaps are disproportionately concentrated in economically weaker territories.

Ethical and Social Implications

The indicators selected in this study carry important ethical and social dimensions that extend beyond their technical definition.

The Wildfire Risk Index concentrates the analysis on the territories of greatest fire danger, ensuring that prevention resources and policy attention are directed where they are most needed. The weighting scheme assigns the greatest importance to conjunctural hazard, reflecting the central role of dynamic climatic and meteorological conditions in determining fire danger, while historical burned areas and critical locations capture structural and recurrent risk patterns. The transparency of this construction allows results to be scrutinised and reproduced, which is itself an ethical requirement of evidence-based public policy.

The Coverage and Gap Ratios translate infrastructure presence into a measurable assessment of protected versus unprotected territory. This framing is analytically useful but it is important to acknowledge that proximity to a fuel management corridor does not guarantee complete protection from wildfire. The use of a 1 km buffer is a transparent and operationally justified threshold, supported by sensitivity analysis at 500 m and 2 km, but communities just outside this boundary may still face significant levels of exposure.

The use of purchasing power as a vulnerability indicator reflects a deliberate choice to foreground economic inequality as a lens for evaluating prevention equity. This is ethically significant: it positions wildfire resilience not merely as a technical or environmental problem, but as a social justice issue in which economically marginalised communities, often rural, elderly, and with limited institutional capacity, may bear a disproportionate share of wildfire risk. At the same time, vulnerability is multi-dimensional, and factors such as population ageing, social isolation, and access to emergency services are not fully captured by an economic index alone. Future analyses could benefit from composite vulnerability indicators that integrate these dimensions more comprehensively.

Finally, the municipal scale of analysis is well suited for policy targeting, as it aligns with the administrative units through which prevention investment and governance decisions are made in Portugal. While intra-municipal variation may exist, the municipal level provides the most actionable and institutionally meaningful unit for identifying priority territories and informing resource allocation decisions within the Smart Cities framework.

2 Data Analytics, Optimization and Policy Suggestions

2.1 Datasets

The analysis integrates nine datasets covering wildfire risk, prevention infrastructure, administrative boundaries, and economic vulnerability. Table 1 provides an overview of all datasets used.

Table 1: Datasets used in the analysis

Dataset	Source	Format
Burned Area Perimeters	ICNF[14]	Shapefile
Conjunctural Hazard Maps	ICNF[14]	GeoTIFF
Critical Fire Locations	ICNF[14]	Shapefile
Vegetation Index (IQV)	ICNF[14]	Shapefile
Aridity Index	ICNF[14]	Shapefile
Primary Fuel Management Network	ICNF[14]	Shapefile
Municipal Boundaries (CAOP)	DGT[15]	Shapefile
Portugal Boundaries	ArcGIS Living Atlas[16]	Shapefile
Municipal Purchasing Power (IpC)	INE[17]	CSV

Burned area perimeters record the spatial extent of fire events across mainland Portugal for each year between 2021 and 2024, totalling 5,998 individual fire perimeters across the four-year period. Conjunctural hazard maps provide annual raster layers of fire danger conditions derived from climatic and meteorological variables. Critical fire locations identify areas of structural susceptibility to wildfire ignition and spread. The vegetation index and aridity index provide spatial layers of fuel load and drought stress respectively, both at the polygon level. All five datasets, together with the Primary Fuel Management Network, were obtained from the Instituto da Conservação da Natureza e das Florestas (ICNF) [14].

Municipal boundaries were sourced from the "Carta Administrativa Oficial de Portugal- CAOP" , published by the Direção-Geral do Território (DGT) [15]. This layer provides the official administrative boundaries for all municipalities in mainland Portugal and serves as the spatial unit for the municipal-level analysis.

Purchasing power data were obtained from the INE 2023 Municipal Purchasing Power Study (*Estudo sobre o Poder de Compra Concelhio*), which provides the per capita income index (IpC) for each municipality relative to the national average [17].

2.2 Data Preparation

Data Cleaning

All datasets were inspected and cleaned before analysis to ensure consistency and correctness. For the conjunctural hazard rasters, nodata values were identified and masked separately for each annual layer, since each yearly file used a different encoding convention: 255 for 2021, no explicit nodata for 2022, and 15 for 2023 and 2024. Zero-value pixels were also excluded, as they indicate areas with no recorded hazard rather than genuinely low risk values. For the social vulnerability data from the INE 2023 Municipal Purchasing Power Study, values were parsed from a semicolon-delimited CSV file encoded in Latin-1, decimal commas were converted to floating-point numbers, and rows with missing values were removed. Municipality names were then normalised through Unicode accent removal and lowercasing to allow consistent matching between the INE dataset and the official administrative boundary file, producing a final set of 278 matched municipalities for the municipal-level analysis.

Spatial Harmonisation

All datasets were aligned to a common spatial reference system, EPSG:3763 (PT-TM06/ETRS89), the official Portuguese national grid, and resampled to a uniform 250 m resolution reference grid. The conjunctural hazard rasters were originally provided at 25 m resolution; to make it computationally feasible to process multiple large national-scale rasters simultaneously, all rasters were downsampled by a factor of 10 using `rasterio`'s out-shape resampling, dividing the number of pixels by 100 while preserving the spatial patterns relevant to the analysis. A mainland Portugal boundary mask was derived from the national administrative boundary shapefile and rasterised onto the reference grid to restrict all outputs to continental territory, excluding the Atlantic archipelagos. All vector layers, burned area perimeters, critical fire locations, vegetation index, aridity index, and the primary fuel management network, were verified against the reference coordinate system prior to rasterisation.

Municipality-Level Integration

Vector layers were converted to raster format at 250 m resolution using the hazard layer as the reference grid, via `rasterio.features.rasterize`. Categorical attributes were mapped to ordinal numerical values prior to rasterisation: the vegetation quality index (*Índice de Qualidade da Vegetação*, IQV) was encoded on a five-point scale from *Baixa* (Low, 1) to *Muito alta* (Very High, 5), and the aridity index from *Húmido* (Humid, 1) to *Semiárido* (Semi-arid, 4). Burned area perimeters for 2021–2024 were merged into a single cumulative binary layer (burned = 5, unburned = 1 on the normalised 1–5 scale). All layers were then normalised to a common 1–5 scale and combined through a weighted overlay to produce the composite Wildfire Risk Index, described in detail in Section ???. Municipal-level statistics were subsequently computed by iterating over each municipality polygon, rasterising its boundary onto the reference grid, and intersecting it with the risk and coverage layers to extract per-municipality totals. Social vulnerability scores were joined to the municipal geodataframe using the normalised municipality names as the common key.

2.3 Wildfire Risk Assessment

The wildfire risk model is constructed through a weighted overlay of five spatial input layers, each normalised to a common 1–5 ordinal scale prior to combination, where 1 represents very low risk and 5 represents very high risk. Conjunctural hazard is derived as the pixel-wise mean of the four annual hazard rasters (2021–2024), normalised to a 0–1 scale before scaling to 1–5. Burned areas are treated as a binary layer: pixels that burned at least once between 2021 and 2024 receive a value of 5 (maximum risk), while unburned pixels receive a value of 1 (minimum risk). Critical fire locations are rasterised as a binary presence layer and subsequently normalised to the 1–5 scale, where pixels coinciding with a critical location receive the maximum value. The vegetation index (IQV) and aridity index are each normalised to 1–5 using min-max normalisation, with zero values treated as no-data and excluded from the calculation.

Figure 1 shows the spatial distribution of burned areas across the four-year study period.

Cumulative Burned Areas in Portugal (2021–2024)

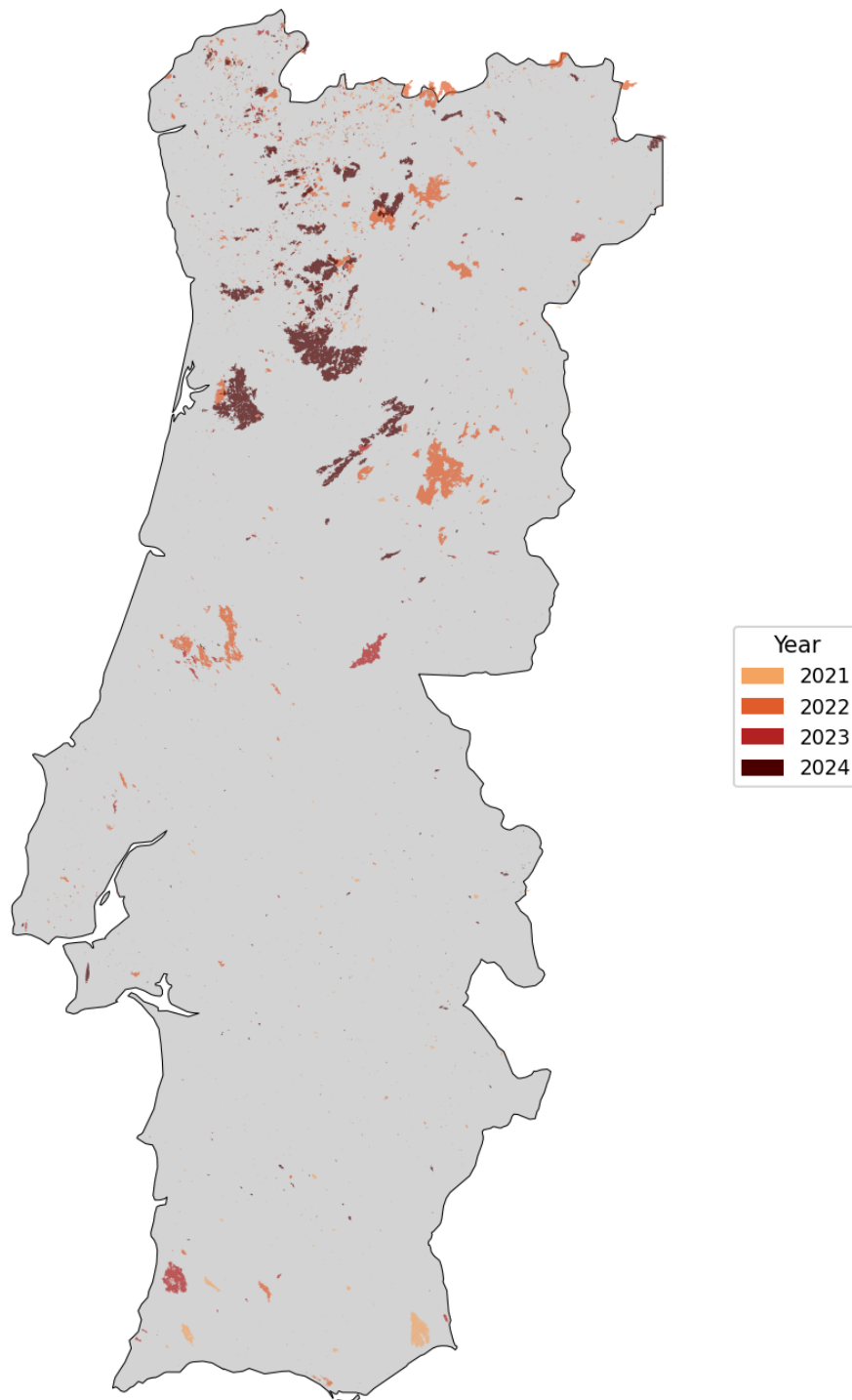


Figure 1: Burned area perimeters across mainland Portugal, 2021–2024.

Figure 2 shows the mean conjunctural hazard across the study period.

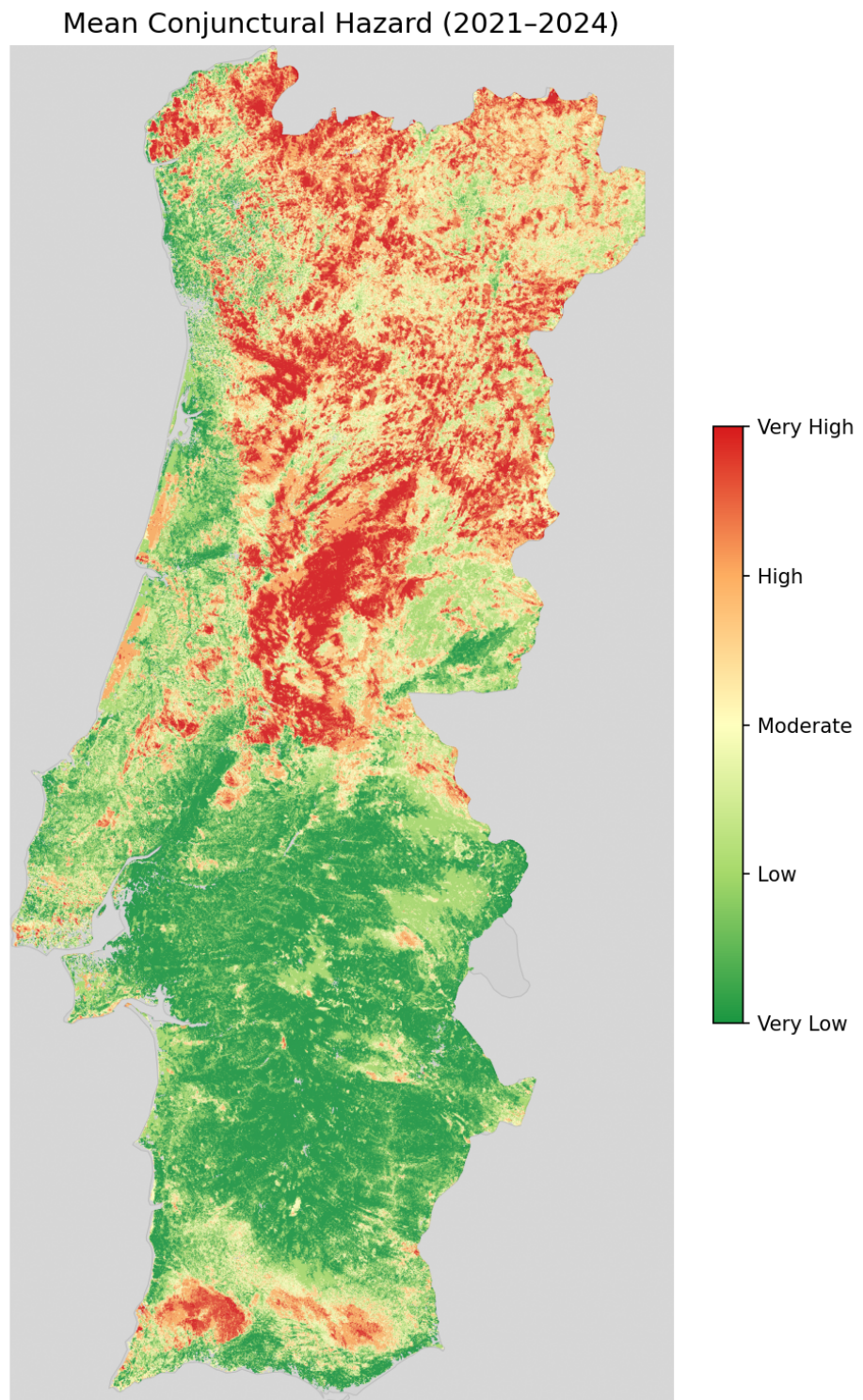


Figure 2: Mean conjunctural hazard index across mainland Portugal, 2021–2024.

Figure 3 shows the spatial distribution of the vegetation index.

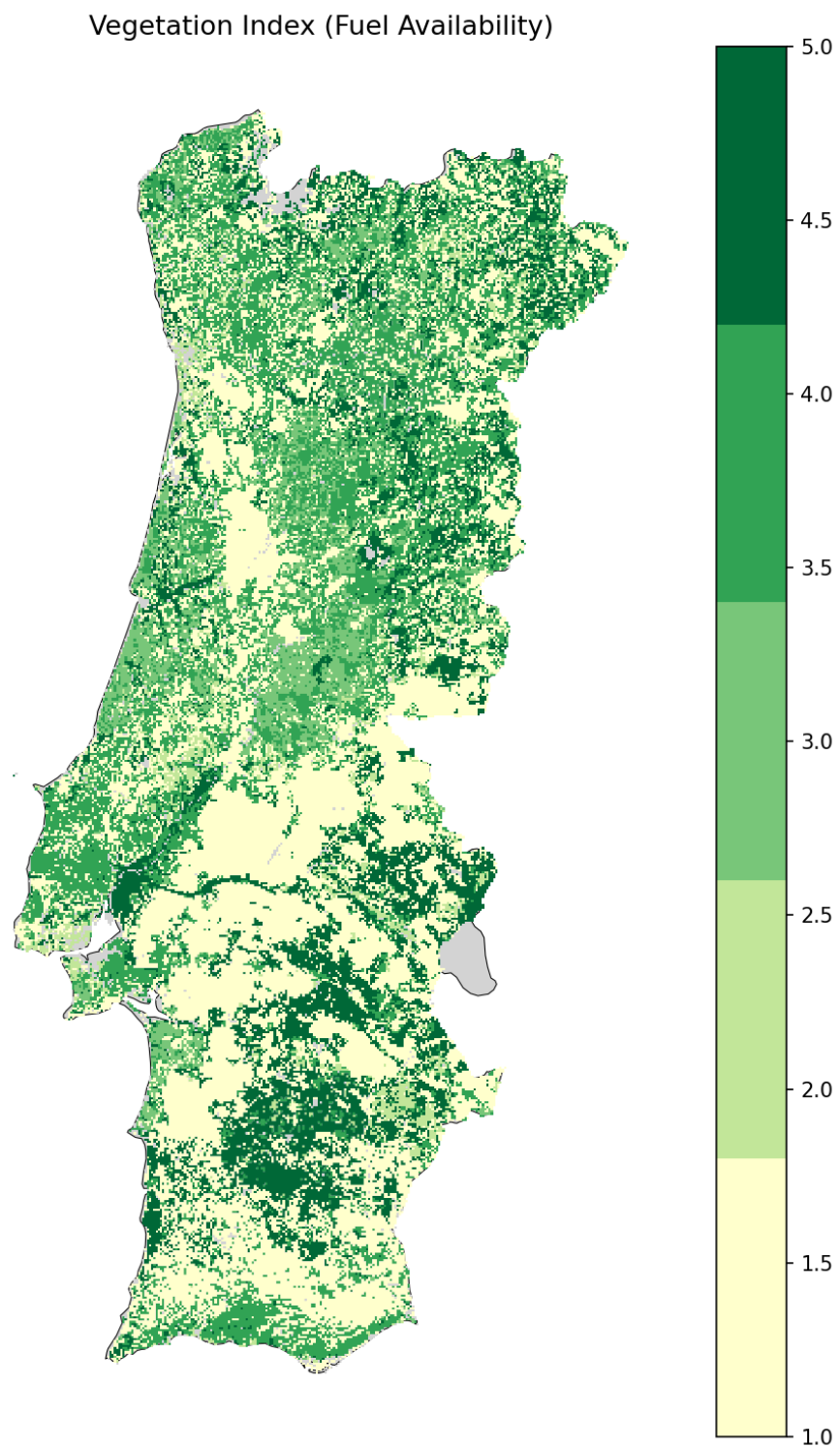


Figure 3: Vegetation index (IQV) across mainland Portugal.

Figure 4 shows the spatial distribution of the aridity index.

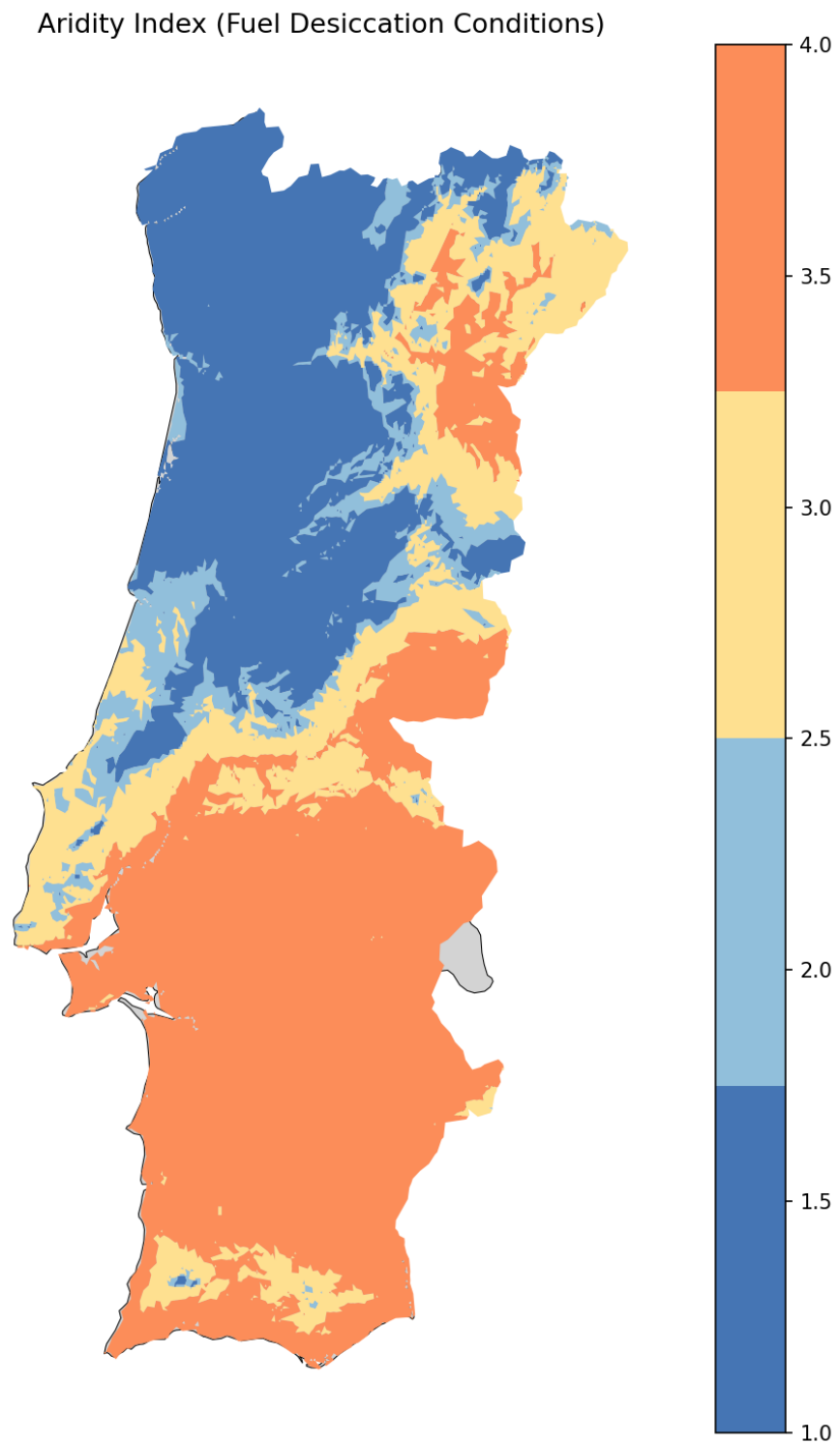


Figure 4: Aridity index across mainland Portugal.

Figure 5 shows the spatial distribution of critical fire locations.

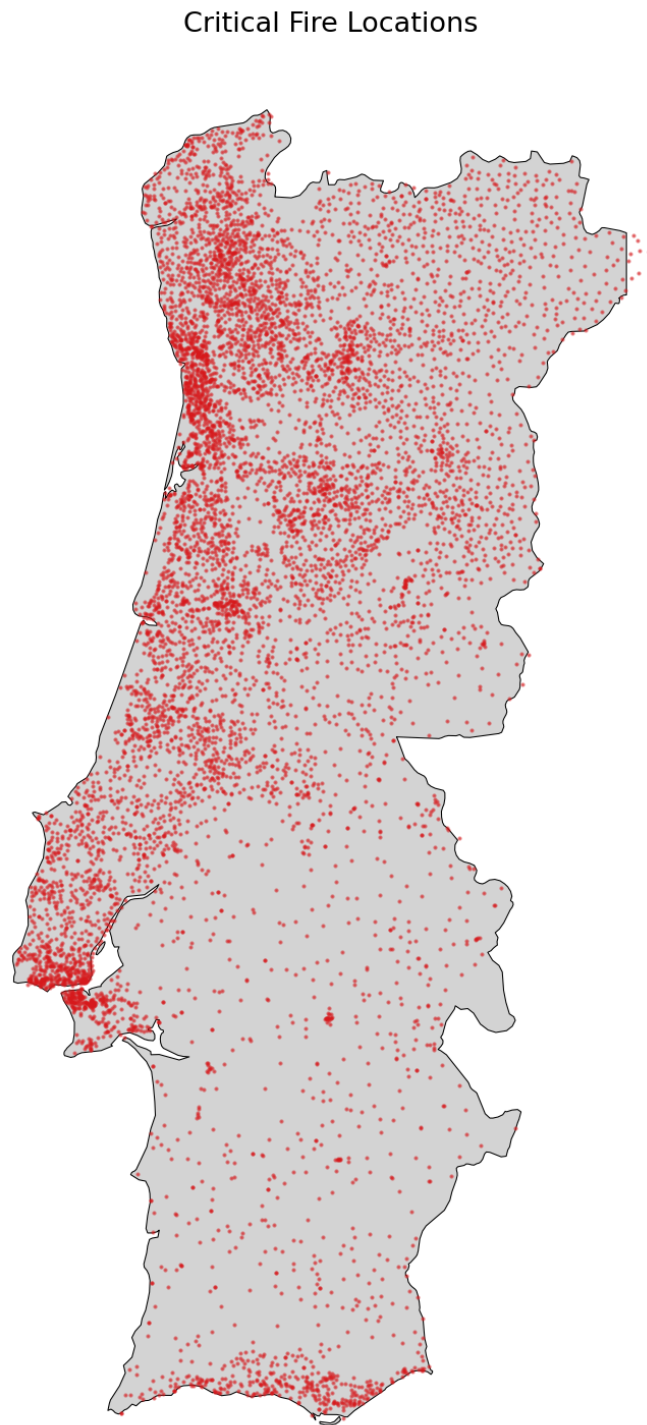


Figure 5: Critical fire locations across mainland Portugal.

The composite risk score is computed as a weighted overlay:

$$\text{Risk} = 0.35 \times \text{Hazard} + 0.20 \times \text{Critical Locations} + 0.20 \times \text{Burned Areas} + 0.15 \times \text{Vegetation} + 0.10 \times \text{Aridity}$$

The weights reflect the relative contribution of each factor to wildfire danger. Conjunctural hazard receives the highest weight (0.35) as it captures the most dynamic and immediate fire danger conditions driven by meteorological variables. Burned areas and critical locations each receive a weight of 0.20, encoding historical fire recurrence and structural territorial susceptibility respectively. Vegetation index (0.15) reflects fuel availability, and aridity (0.10) encodes background climatic stress. Where a layer contains no-data for a given pixel, the remaining layers are renormalised by their combined weight, ensuring that missing data does not bias the composite score. The resulting continuous surface is subsequently discretised into five risk classes using equal interval classification.

Figure 6 shows the final national wildfire risk map.

Integrated Wildfire Risk Model — Portugal (2021–2024)

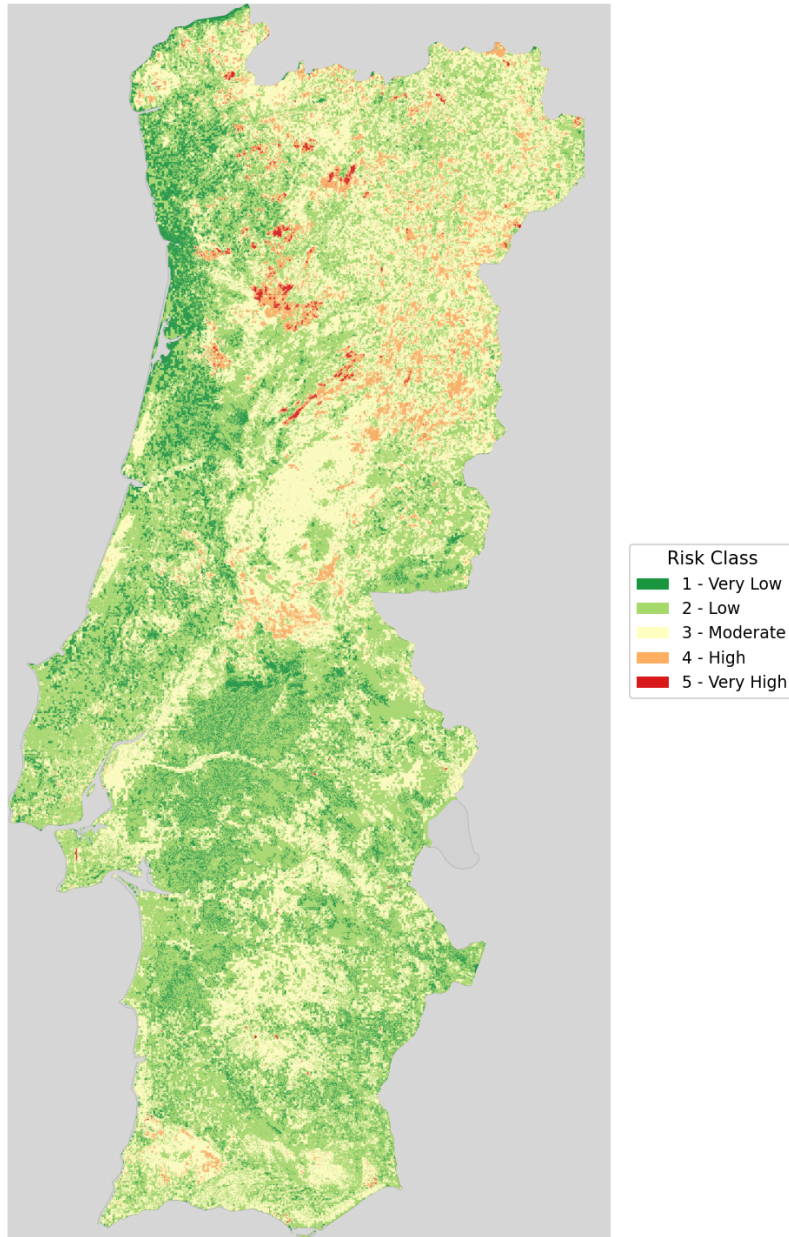


Figure 6: Integrated Wildfire Risk Map for mainland Portugal (2021–2024). Risk classes range from 1 (Very Low) to 5 (Very High).

Identification of High-Risk Areas

Following classification, only areas assigned to risk class 4 (High) or class 5 (Very High) are retained for the coverage analysis, as these represent the territories where the absence of prevention infrastructure carries the greatest potential consequences. Each raster pixel covers an area of 0.0625 km^2 ($250 \text{ m} \times 250 \text{ m}$), allowing direct conversion of pixel counts to areas in km^2 .

Figure 7 shows the total area assigned to each risk class.

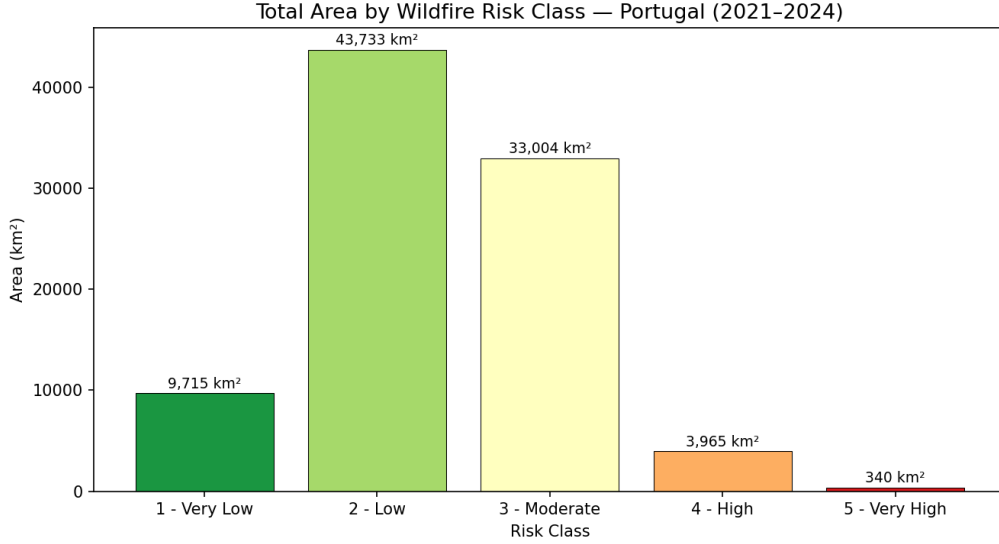


Figure 7: Total area by wildfire risk class across mainland Portugal (2021–2024). Classes 4 and 5 form the basis of all subsequent coverage analysis.

2.4 Prevention Coverage Analysis

To assess whether a high-risk area is covered by the Primary Fuel Management Network (RPFGC), a zone of influence is defined around the network’s linear infrastructure using a raster buffer approach. The fuel network is first rasterised onto the same 250 m reference grid used for the risk model, assigning a value of 1 to pixels intersecting a network segment and 0 elsewhere. Buffer zones are then computed via binary dilation of the rasterised network, expanding the covered area by a defined number of pixels in all directions. At 250 m resolution, 2 pixels correspond to 500 m, 4 pixels to 1 km, and 8 pixels to 2 km.

Three buffer distances are tested, 500 m, 1 km, and 2 km, to assess the sensitivity of coverage results to this methodological choice. A high-risk pixel (class 4 or 5) is classified as *covered* if it falls within the buffer zone, and as *uncovered* otherwise. The 1 km buffer is adopted as the primary threshold for the municipal-level analysis, as it represents a operationally meaningful distance within which fuel management infrastructure can reasonably be expected to reduce fire spread. Figure 8 shows the spatial distribution of the Primary Fuel Management Network across mainland Portugal.

Primary Fuel Management Network

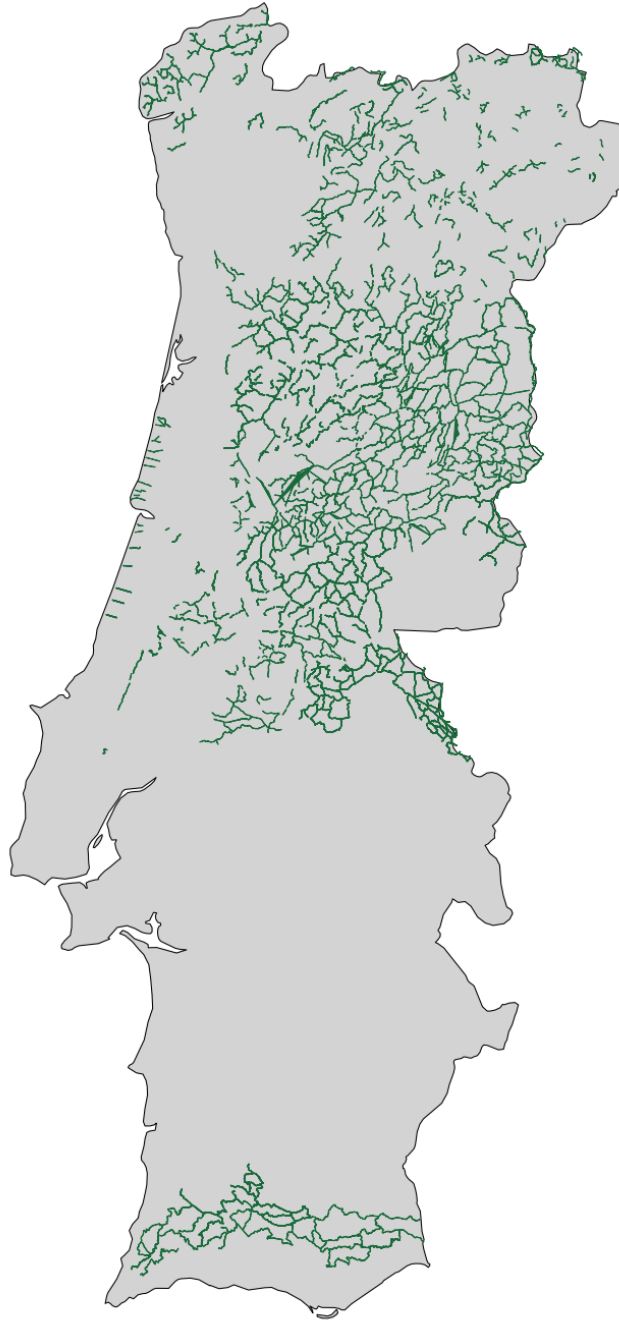


Figure 8: Primary Fuel Management Network (RPFGC) across mainland Portugal.

Figure 9 shows the spatial overlap between the integrated wildfire risk model and the Primary Fuel Management Network. The map allows a direct visual assessment of the alignment between prevention infrastructure and high-risk zones, highlighting areas where the network is present within high-risk territory and, conversely, where high-risk areas remain without network coverage.

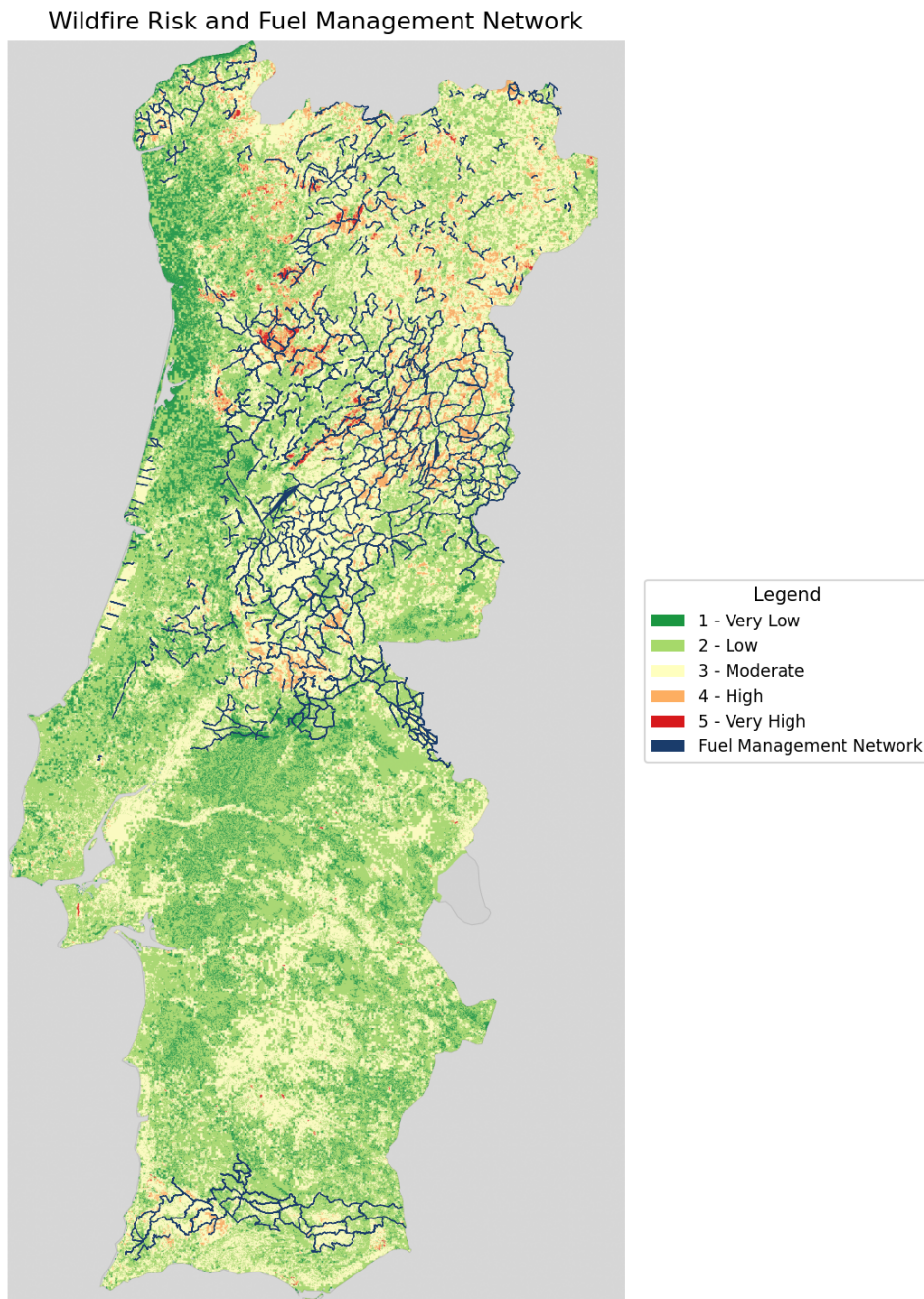


Figure 9: Spatial overlay of the integrated wildfire risk model and the Primary Fuel Management Network across mainland Portugal (2021–2024). Risk classes range from 1 (Very Low) to 5 (Very High). The network is shown in dark blue.

Coverage Assessment and National Coverage Statistics

For each buffer distance, the spatial intersection between the buffer zone and the high-risk area (classes 4 and 5) is computed, and the covered and uncovered areas are calculated by multiplying pixel counts by the pixel area of 0.0625 km². Table 2 summarises the national coverage statistics for each buffer distance.

Table 2: National coverage statistics for high-risk areas (classes 4–5) by buffer distance.

Buffer	Total HR Area (km ²)	Covered (km ²)	Uncovered (km ²)	Coverage (%)
500 m	4,326.56	940.06	3,386.50	21.7
1 km	4,326.56	1,670.75	2,655.81	38.6
2 km	4,326.56	2,660.50	1,666.06	61.5

At the primary 1 km threshold, the Coverage Ratio is 38.6%, meaning that approximately 1,671 km² of the 4,327 km² of high-risk territory falls within the zone of influence of the network, while 61.4%, corresponding to approximately 2,656 km², remains unprotected. Even at the most generous 2 km buffer, 38.5% of high-risk territory remains outside the network’s zone of influence. These results indicate that, at the national scale, a substantial proportion of high wildfire risk territory in mainland Portugal is not covered by the Primary Fuel Management Network. Figure 10 illustrates the covered and uncovered areas for each buffer distance.

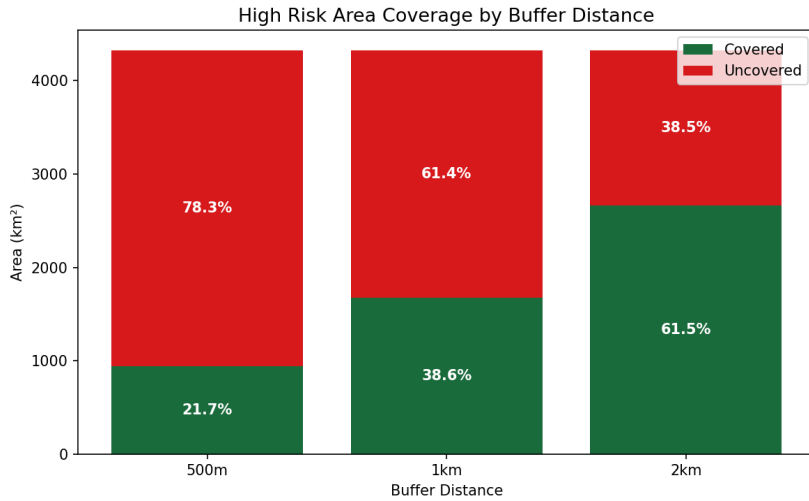


Figure 10: High-risk area coverage by buffer distance. Green bars represent covered area and red bars represent uncovered area (km²). Percentages indicate the proportion of total high-risk area in each category.

2.5 Municipal-Level Analysis

Coverage Ratio and Gap Ratio by Municipality

To identify which municipalities face the greatest prevention gaps, coverage statistics are computed at the municipal level using the 1 km buffer as the primary analysis distance. For each municipality, the high-risk area (classes 4–5), the covered portion, the uncovered portion, the Coverage Ratio, and the Gap Ratio are calculated by spatially intersecting the municipal boundaries with the high-risk mask and the buffer zone. Municipalities with a high-risk area below 1 km² are excluded from the ranking to avoid results dominated by very small territorial units. Municipalities shown in grey in the map have no high-risk area within their boundaries and are therefore not part of the coverage analysis, as there is no high-risk territory to measure coverage against.

Figure 11 shows the Gap Ratio map by municipality and the top 10 municipalities by absolute uncovered high-risk area.

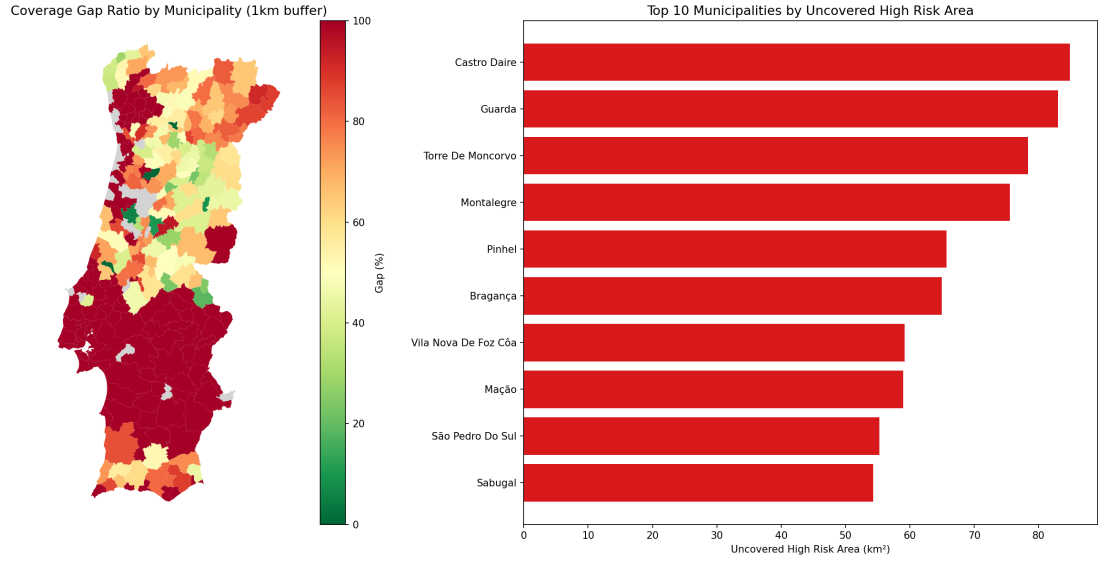


Figure 11: Left: Coverage Gap Ratio by municipality (1 km buffer). Right: Top 10 municipalities by uncovered high-risk area (km²).

The municipalities with the largest absolute uncovered high-risk areas are concentrated in the northern and central interior of Portugal: Castro Daire, Guarda, Torre de Moncorvo, Montalegre, and Bragança among them. These are predominantly rural, inland municipalities characterised by high wildfire exposure, low population density, and limited institutional capacity, making the scale of unprotected high-risk territory particularly concerning.

2.6 Economic Vulnerability Analysis

Purchasing Power Distribution

Economic vulnerability at the municipal level is measured using the per capita income index (IpC) from the INE 2023 Municipal Purchasing Power Study (*Estudo sobre o Poder de Compra Concelhio*) [17]. The IpC expresses the purchasing capacity of each municipality's resident population relative to the national average, which is set at a base value of 100. Municipalities with an IpC below 100 have below-average purchasing power and are considered economically vulnerable for the purposes of this study.

The INE dataset is loaded and matched to the municipal boundaries shapefile by municipality name. To account for differences in accentuation and capitalisation between the two sources, municipality names are normalised by removing accents and converting to lowercase prior to matching.

Table 3 shows the ten most economically vulnerable municipalities in mainland Portugal, ranked by their vulnerability score. All ten municipalities have an IpC well below the national average of 100, with Tabuaço recording the lowest value (IpC = 63.53, Vulnerability = 1.00), followed closely by Penamacor, Vimioso, and Vinhais. These are predominantly small, rural interior municipalities with significantly below-average purchasing power.

Table 3: Top 10 most economically vulnerable municipalities, ranked by vulnerability score (INE 2023).

Municipality	IpC	Vulnerability Score
Tabuaço	63.53	1.000
Penamacor	63.77	0.998
Vimioso	63.80	0.998
Vinhais	64.26	0.994
Baião	64.54	0.991
Sernancelhe	64.72	0.990
Melgaço	64.90	0.988
Valpaços	64.93	0.988
Santa Marta de Penaguião	65.00	0.988
Celorico de Basto	65.08	0.987

Economic Vulnerability Mapping

The IpC is inverted and normalised to a 0–1 vulnerability score, where 1 represents the most economically vulnerable municipality and 0 represents the least vulnerable:

$$\text{Vulnerability Score} = 1 - \frac{\text{IpC}_i - \min(\text{IpC})}{\max(\text{IpC}) - \min(\text{IpC})}$$

This transformation ensures that all subsequent analyses use a consistent and comparable scale across municipalities. Figure 12 shows the spatial distribution of vulnerability scores across mainland Portugal.

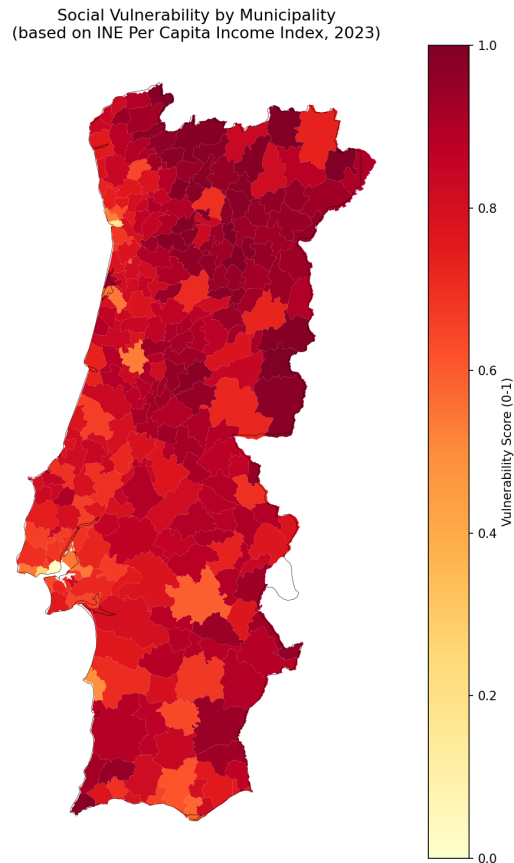


Figure 12: Economic vulnerability score by municipality, based on the INE 2023 per capita income index (IpC). Darker colours indicate higher vulnerability.

2.7 Relative and Absolute Coverage Gaps and Economic Vulnerability

The relationship between prevention coverage gaps and economic vulnerability is central to the equity dimension of this study. If municipalities with lower purchasing power systematically exhibit higher Gap Ratios, this would suggest that the allocation of wildfire prevention infrastructure is spatially inequitable, leaving the most economically fragile territories also the least protected. Such a pattern would have significant implications for public policy, as it would indicate that communities with the least capacity to self-protect are also those receiving the least institutional support. Conversely, if no such relationship exists, it would suggest that the network is broadly risk-informed rather than socially biased. To provide a comprehensive assessment, two complementary analyses are conducted: one examining the *relative* coverage gap: the percentage of each municipality's high-risk area that is unprotected, and one examining the *absolute* unprotected area: the total km² of high-risk land without network coverage. Together, these two perspectives allow for a more complete understanding of territorial inequalities in wildfire resilience.

Correlation Analysis

To examine this relationship, the municipal coverage statistics computed using the 1 km buffer are joined with the vulnerability scores derived from the INE 2023 IpC index. The two indicators: Gap Ratio and Vulnerability Score, are then compared through a scatter plot and a Pearson correlation analysis.

Figure 13 shows the scatter plot of Vulnerability Score against Gap Ratio for all municipalities with high-risk area. Each point represents one municipality, coloured by the size of its total high-risk area. A linear trend line is fitted to the data to assess the direction and strength of the relationship.

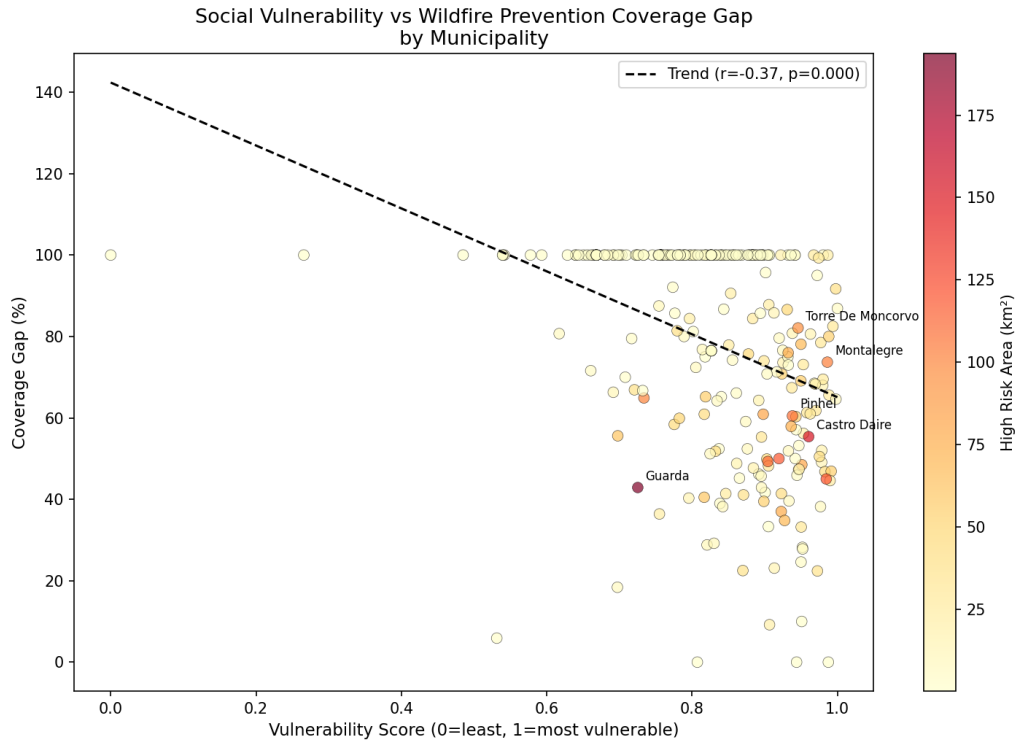


Figure 13: Scatter plot of economic vulnerability score versus coverage gap ratio by municipality. Each point represents one municipality, coloured by total high-risk area (km²). The dashed line shows the linear trend.

The Pearson correlation between vulnerability score and Gap Ratio is $r = -0.37$ ($p < 0.001$), indicating a statistically significant negative relationship between economic vulnerability and prevention coverage gaps at the municipal level. The results of this analysis are discussed in detail in Section 2.8.

Figure 14 complements the previous analysis by examining the relationship between economic vulnerability and the absolute area of unprotected high-risk territory per municipality. Unlike Figure 13, which shows the

relative Gap Ratio, this plot reveals the absolute scale of the problem. The Pearson correlation is $r = 0.30$ ($p < 0.001$), indicating a statistically significant positive relationship: more economically vulnerable municipalities tend to have larger absolute unprotected high-risk areas. The darkest points concentrate in the upper right corner, representing large, poor, interior municipalities such as Castro Daire, Guarda, Torre de Moncorvo, and Montalegre, which combine high economic vulnerability with the largest absolute prevention gaps.

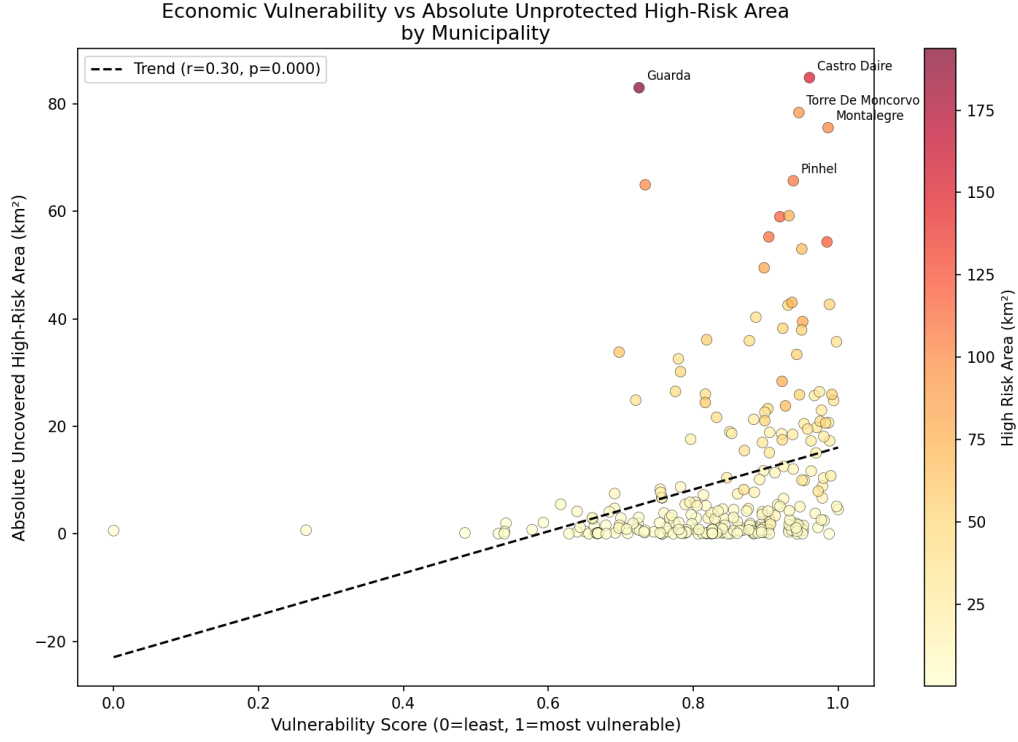


Figure 14: Scatter plot of economic vulnerability score versus absolute uncovered high-risk area (km^2) by municipality. Point colour indicates total high-risk area size. The dashed line shows the linear trend ($r = 0.30$, $p < 0.001$).

Table 4 presents the ten municipalities with the largest absolute uncovered high-risk areas, combining coverage statistics with their economic vulnerability scores. The municipalities are ranked by total uncovered area in descending order. For each municipality, the table reports the total high-risk area (km^2), the Coverage Ratio, the Gap Ratio, and the Vulnerability Score. This combined view allows a direct comparison between the scale of unprotected high-risk territory and the economic capacity of each municipality to respond to wildfire risk.

Table 4: Top 10 municipalities by uncovered high-risk area, with coverage statistics and economic vulnerability score.

Municipality	High Risk Area (km^2)	Coverage (%)	Gap (%)	Vulnerability Score
Castro Daire	153.31	44.6	55.4	0.96
Guarda	193.69	57.1	42.9	0.73
Torre de Moncorvo	95.44	17.9	82.1	0.95
Montalegre	102.56	26.3	73.7	0.99
Pinhel	108.50	39.5	60.5	0.94
Bragança	100.06	35.1	64.9	0.73
Vila Nova de Foz Côa	77.88	24.0	76.0	0.93
Mação	118.06	50.0	50.0	0.92
São Pedro do Sul	112.12	50.7	49.3	0.90
Sabugal	120.69	55.0	45.0	0.98

Figure 15 shows the correlation heatmap between all key municipal indicators.

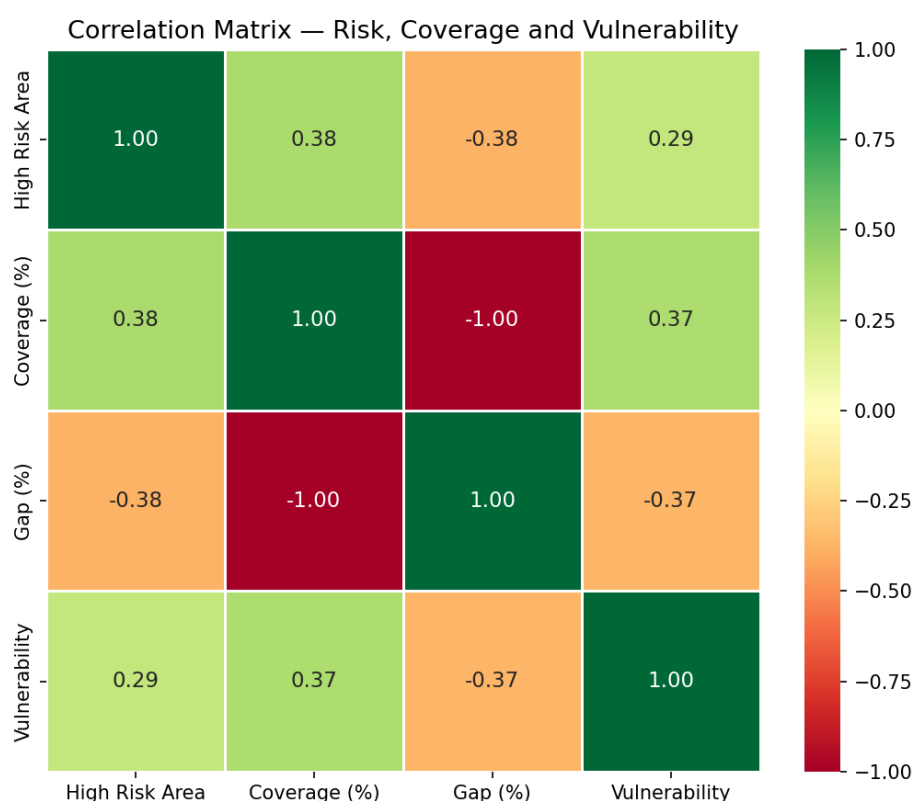


Figure 15: Correlation heatmap between municipal indicators: high-risk area, coverage ratio, gap ratio, and vulnerability score.

The heatmap confirms that coverage ratio and gap ratio are perfectly inversely correlated by construction. Notably, municipalities with larger high-risk areas tend to have slightly better coverage ($r = 0.38$), and more vulnerable municipalities also tend to have marginally better coverage ($r = 0.37$), consistent with the finding that the network is concentrated where risk and vulnerability overlap. However, these correlations also reveal that a significant share of high-risk, high-vulnerability territory remains unprotected in absolute terms, pointing to the need for targeted expansion of the network in the most critical municipalities.

Priority Classification

To further contextualise the relationship between coverage gaps and economic vulnerability, municipalities are classified into five priority levels for prevention investment. The classification combines three criteria: the Gap Ratio, the vulnerability score, and the total high-risk area. Only municipalities with a high-risk area of at least 10 km² are included. The median Gap Ratio and median vulnerability score are used as thresholds to assign each municipality to a priority level, as follows:

- **Very High:** Gap Ratio above median and vulnerability above median
- **High:** Gap Ratio above median or vulnerability above median, but not both
- **Medium:** Gap Ratio below median and vulnerability above median
- **Low-Medium:** Gap Ratio above median and vulnerability below median
- **Low:** Gap Ratio below median and vulnerability below median

Figure 16 shows the resulting priority map and the top priority municipalities by uncovered high-risk area.

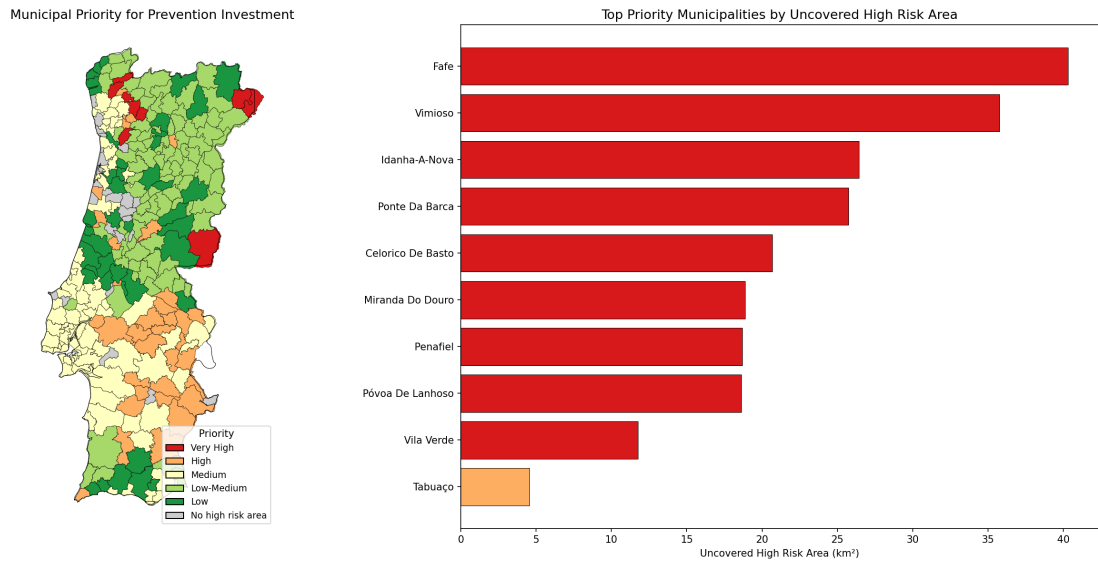


Figure 16: Left: Municipal priority classification for prevention investment. Right: Top priority municipalities by uncovered high-risk area (km²).

The median Gap Ratio threshold is 86.8% and the median vulnerability threshold is 0.851. Table 5 summarises the resulting distribution of municipalities across priority levels, and Table 6 lists the 9 Very High priority municipalities.

Table 5: Distribution of municipalities by priority level.

Priority Level	Number of Municipalities
Very High	9
High	35
Medium	92
Low-Medium	92
Low	44
Total	272

Table 6: Very High priority municipalities with key indicators.

Municipality	Gap (%)	Vulnerability Score	High Risk Area (km ²)
Penafiel	90.6	0.853	20.62
Fafe	100.0	0.887	40.31
Celorico de Basto	100.0	0.987	20.69
Ponte da Barca	100.0	0.967	25.75
Vila Verde	100.0	0.897	11.75
Póvoa de Lanhoso	100.0	0.922	18.62
Idanha-a-Nova	99.3	0.974	26.62
Vimioso	91.7	0.998	39.00
Miranda do Douro	87.8	0.906	21.50

2.8 Results and Discussion

Coverage Effectiveness

The integrated wildfire risk model identifies a total high-risk area (classes 4 and 5) of approximately 4,327 km² across mainland Portugal, corresponding to roughly 4.7% of the national territory. Class 4 (High) accounts for 3,965 km² and Class 5 (Very High) for 361 km², while the majority of the territory falls within lower risk classes (Figure 7). As shown in Figure 7, classes 4 and 5 together represent only 4.7% of the national territory, while classes 1 and 2 account for over 75%, indicating that the highest risk levels affect a relatively small but critical portion of mainland Portugal.

At the national scale, only 38.6% of high-risk territory falls within the 1 km zone of influence of the Primary Fuel Management Network, leaving 61.4%, approximately 2,656 km², unprotected (Table 2, Figure 10). Even at the most generous 2 km buffer, 38.5% of high-risk territory remains outside the network's zone of influence. These results indicate that, at the national scale, the current deployment of the RPFGC covers less than half of the territory most exposed to wildfire risk, representing a significant gap in prevention infrastructure. This finding is consistent with the literature highlighting low execution rates and insufficient targeting in Portuguese fuel management policy [12].

Economic Vulnerability and Prevention Gaps

The economic vulnerability analysis reveals that mainland Portugal is predominantly characterised by below-average purchasing power, particularly in the northern and central interior (Figure 12). The most economically vulnerable municipalities include Tabuaço (IpC = 63.53, Vulnerability = 1.00), Penamacor (IpC = 63.77, Vulnerability = 1.00), Vimioso (IpC = 63.80, Vulnerability = 1.00), and Vinhais (IpC = 64.26, Vulnerability = 0.99), all of which are rural interior municipalities with significantly below-average economic capacity.

The Pearson correlation between vulnerability score and Gap Ratio is $r = -0.37$ ($p < 0.001$), indicating a statistically significant negative relationship (Figure 13). In the scatter plot, each point represents one municipality, with the X axis showing the vulnerability score and the Y axis showing the Gap Ratio. The dashed trend line slopes downward from left to right, meaning that as vulnerability increases, the Gap Ratio tends to decrease. Contrary to a simple inequality hypothesis, more economically vulnerable municipalities do not systematically exhibit larger relative coverage gaps. This is explained by the spatial concentration of the RPFGC in the northern and central interior, precisely where both wildfire risk and economic vulnerability are highest, suggesting that network placement is broadly risk-informed rather than socially biased.

However, this result must be interpreted alongside the absolute dimension of coverage gaps. A complementary analysis of the relationship between vulnerability and absolute uncovered area yields a positive correlation of $r = 0.30$ ($p < 0.001$) (Figure 14), indicating that more vulnerable municipalities carry significantly larger absolute unprotected areas. The two correlations together tell a complete story: the network broadly follows risk patterns in relative terms, but the absolute burden of unprotected high-risk territory falls disproportionately on the most economically fragile municipalities. As shown in Table 4, the municipalities with the largest absolute uncovered high-risk areas: Castro Daire (153 km², Gap = 55.4%, Vulnerability = 0.96), Guarda (193 km², Gap = 42.9%, Vulnerability = 0.73), Torre de Moncorvo (95 km², Gap = 82.1%, Vulnerability = 0.95), and Montalegre (102 km², Gap = 73.7%, Vulnerability = 0.99), are all highly vulnerable interior municipalities. The network is present in these territories, but it is not large enough to cover the scale of risk that exists there. Since these municipalities also lack the economic capacity to self-protect, the result is a territorial inequality that is absolute rather than relative in nature.

Territorial Inequalities in Wildfire Resilience

The correlation heatmap (Figure 15) confirms that municipalities with larger high-risk areas tend to have slightly better relative coverage ($r = 0.38$), and more vulnerable municipalities also tend to have marginally better coverage ($r = 0.37$). While this suggests the network broadly follows risk and vulnerability patterns, it also reveals that a significant share of high-risk, high-vulnerability territory remains unprotected in absolute terms.

The spatial distribution of Gap Ratios (Figure 11) reveals a clear geographic pattern: municipalities with the largest proportions of unprotected high-risk territory are predominantly located in the northern and central interior of Portugal, a region characterised by extensive forested areas, high wildfire exposure, and low population density. Coastal and southern municipalities tend to exhibit lower Gap Ratios, reflecting either lower wildfire risk or better relative coverage by the network.

Taken together, these results suggest that while the RPFGC is broadly aligned with territorial risk, its current scale is insufficient to protect the most exposed and vulnerable municipalities. A municipality with over

80 km² of uncovered high-risk area and a vulnerability score above 0.95 faces a fundamentally different challenge than a coastal municipality with less than 1 km² uncovered, even if their relative Gap Ratios appear similar. This distinction between relative and absolute inequality is central to understanding the territorial resilience challenges facing Portugal's interior municipalities.

The priority classification identifies 9 municipalities as Very High priority, 35 as High priority, 92 as Medium, 92 as Low-Medium, and 44 as Low priority (Figure 16). The 9 Very High priority municipalities combine above-median Gap Ratios, above-median vulnerability scores, and large high-risk areas, representing the most urgent cases for targeted prevention investment. The fact that only 9 municipalities fall into the Very High category reflects the concentration of the worst-case combination, high risk, low coverage, and high vulnerability, in a small number of interior territories, reinforcing the case for geographically targeted rather than uniform national intervention.

These findings are broadly consistent with the existing literature on wildfire risk and prevention in Portugal. Benali et al. [12] highlight the importance of spatially targeted fuel management interventions at the local scale, arguing that prevention resources must be concentrated where structural risk is highest [12]. The OECD has similarly identified the need for risk-informed resource allocation and improved prevention infrastructure in Portugal's most exposed territories [7]. The negative correlation between vulnerability and gap ratio found in this study partially supports the hypothesis that the network follows risk patterns, but the absolute scale of unprotected territory in high-vulnerability municipalities points to a need for substantial expansion of prevention infrastructure, consistent with the PRR investment targets [11].

2.9 Policy Interventions and Ethical Implications

The findings of this study point to a set of concrete and evidence-based policy interventions to address the identified gaps in wildfire prevention coverage across mainland Portugal. Each recommendation is grounded in the analytical results and accompanied by a discussion of its ethical and social implications.

Targeted Expansion of the Primary Fuel Management Network

At the national scale, the RPFGC covers only 38.6% of high-risk territory at the 1 km buffer threshold, leaving approximately 2,656 km² unprotected. The 9 Very High priority municipalities identified in this study: Fafe, Vimioso, Idanha-a-Nova, Ponte da Barca, Celorico de Basto, Miranda do Douro, Póvoa de Lanhoso, Vila Verde, and Penafiel, should be the immediate focus of prevention investment. These municipalities combine Gap Ratios above 87%, vulnerability scores above 0.85, representing the worst-case combination of exposure, insufficient coverage, and limited economic capacity to self-protect. The 35 High priority municipalities should constitute the second tier of intervention.

The PRR project C08-i03 targets the establishment of 21,000 hectares of fuel management strips [11], but the results of this study suggest that this investment must be spatially targeted to maximise its impact. Expansion should prioritise municipalities with Gap Ratios above 90% such as Fafe, Celorico de Basto, Ponte da Barca, Vila Verde, and Póvoa de Lanhoso, where the network is virtually absent from high-risk zones.

From an ethical standpoint, prioritising investment based on a combination of risk exposure and economic vulnerability is a redistributive choice that explicitly acknowledges the unequal capacity of municipalities to respond to wildfire risk. This approach is consistent with principles of environmental justice and territorial cohesion, and reflects the responsibility of the state to direct public resources where they are most needed. Transparent reporting of coverage gaps at the municipal level, as demonstrated in this study, is a prerequisite for accountable governance.

Risk-Informed Resource Allocation

The results demonstrate that the current network is broadly risk-informed, it is concentrated where fire risk and vulnerability are highest. However, the absolute scale of unprotected territory in the most exposed municipalities reveals that risk-informed placement alone is insufficient without adequate investment in network density and extent. Resource allocation decisions should therefore be guided by both the Gap Ratio and the absolute uncovered area, ensuring that municipalities with large high-risk territories receive proportionally greater investment regardless of their relative coverage performance.

The purchasing power index used in this study provides a practical proxy for identifying municipalities with limited institutional capacity, but future resource allocation frameworks should consider broader vulnerability indicators including population ageing, housing informality, and access to emergency services, to ensure that no dimension of social fragility is overlooked.

Smart Monitoring and Decision Support Systems

Within the Smart Cities framework, the geospatial and data-driven methodology developed in this study can serve as the basis for a continuous monitoring and decision support system for wildfire prevention policy in Portugal. By regularly updating the risk model, coverage analysis, and vulnerability scores with new data, policymakers can track the evolution of coverage gaps over time, evaluate the impact of network expansion investments, and identify emerging priority territories before they become critical. Integration with real-time sensor data, satellite imagery, and climate projections would further enhance the predictive capacity of such a system [13].

The deployment of such systems raises important ethical considerations around data governance, transparency, and algorithmic accountability. Decisions informed by automated models must remain subject to democratic scrutiny, and the communities most affected by wildfire risk should have access to the data and analyses that shape prevention policy in their territories.

2.10 Conclusion

This study set out to answer the following research question: *How effectively does Portugal’s primary fuel management infrastructure cover high wildfire risk zones, are the gaps in prevention coverage disproportionately concentrated in the most economically vulnerable municipalities, and what targeted policy interventions could address these inequalities?*

The analysis shows that the Primary Fuel Management Network covers only 38.6% of high wildfire risk territory at the 1 km threshold, leaving approximately 2,656 km² unprotected. In relative terms, the evidence does not support the hypothesis that coverage gaps are disproportionately concentrated in the most economically vulnerable municipalities ($r = -0.37$, $p < 0.001$), indicating that the network broadly follows risk patterns. However, in absolute terms, more vulnerable municipalities carry significantly larger unprotected areas ($r = 0.30$, $p < 0.001$), revealing a territorial inequality that is real but absolute rather than relative in nature. The absolute burden of unprotected high-risk territory falls disproportionately on the most economically fragile interior municipalities, which also lack the capacity to self-protect.

In response to the third component of the research question, the study identifies 9 Very High priority municipalities: Fafe, Vimioso, Idanha-a-Nova, Ponte da Barca, Celorico de Basto, Miranda do Douro, Póvoa de Lanhoso, Vila Verde, and Penafiel, as the most urgent targets for targeted expansion of the RPFGC. A risk-informed, equity-conscious, and data-driven approach to prevention investment, supported by smart monitoring systems, represents the most effective path towards reducing wildfire risk and territorial inequality in mainland Portugal.

CRediT Author Statement

All authors contributed to the development of this project. The specific contributions of each author are detailed below according to the CRediT (Contributor Roles Taxonomy) framework.

Inês Maneta: Software, Formal Analysis, Investigation, Writing – Original Draft, Resources.

João Sousa: Software, Validation, Writing – Original Draft, Writing – Review & Editing.

Marta Dias: Software, Data Curation, Visualization, Writing – Original Draft.

Rita Pinto: Conceptualization, Methodology, Software, Project Administration, Writing – Original Draft.

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